

Boost.Int128: A C++14 Library for Portable and Performant 128-Bit Integer Arithmetic

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Summary

Boost.Int128 is a header-only, C++14 implementation of 128-bit integer arithmetic through two types, `int128` and `uint128`, that behave like built-in integers and integrate with the C++ Standard Library and other Boost libraries (Boost Authors, 2026). Both types are exactly 128 bits wide on every supported platform, and both are usable unchanged in GPU kernels through CUDA (Nickolls et al., 2008) and SYCL (Reyes et al., 2020). The library is available at <https://github.com/boostorg/int128>, and in the Boost distribution starting with version 1.93.

Statement of need

C++ specifies no integer type wider than `long long`, which in practice means 64 bits, yet quantities exceeding that range are common. An IPv6 address (Hinden & Deering, 2006) and a UUID (Davis et al., 2024) are each 128 bits, so each needs a 128-bit integer or a byte array that cannot be compared, masked, or incremented directly. The widening 64x64 to 128-bit product is the central primitive of unbiased random integer generation in an interval (Lemire, 2019) and of shortest round-trip floating-point formatting (Adams, 2018). GCC and Clang expose `__int128` only on 64-bit targets, with no `std::numeric_limits`, `<iostream>`, or `std::hash` support, and MSVC lacks it entirely, so dependent code either restricts its portability or maintains a second implementation. The portable alternative, a general multiprecision integer, pays for that generality in object size and in throughput. The types began as the arithmetic backend of Boost.Decimal (Borland & Kormanyos, 2026), which needs an exact 128-bit product on every platform Boost supports, and were extracted and generalized once that need proved broader than one library. What is needed is a type that is exactly 16 bytes everywhere, as fast as the compiler's extension where that exists, and usable across the host-to-device boundary without an interchange format.

State of the field

The portable host implementations are Abseil's `absl::int128` and `absl::uint128` (Google, 2017), Boost.Multiprecision's 128-bit `cpp_int` instantiation (Kormanyos & Maddock, 2026), and MSVC's undocumented `std::_Signed128` and `std::_Unsigned128` (Microsoft Corporation, 2026). Abseil is a close functional match, but it requires C++17 and GCC 10 or later and carries no device annotations. Boost.Multiprecision's types instantiate a general number template rather than a purpose-built class, so they carry an extra word of bookkeeping and cost more per operation.

On the device side, `nvcc` accepts `__int128` only when the host compiler supports it (NVIDIA Corporation, 2026), which excludes every MSVC and 32-bit host, and SYCL's `spir64` target has no native 128-bit integer at all (Khronos SYCL Working Group, 2023). The implementations

40 filling this gap, such as CUDA-uint128 (Seizert, 2016), are unsigned only and tied to one
 41 toolchain. We are not aware of another implementation offering one 128-bit type usable on
 42 the host and in both CUDA and SYCL device code with bitwise identical results.

43 Software design

44 The library requires only C++14, is header-only, and has no external dependencies, so a
 45 user need only `#include <boost/int128.hpp>`, even with a toolchain as old as g++-5. A
 46 single-header amalgamation is published, and under C++20 the library can be consumed
 47 as a module via `import boost.int128`. Relaxed C++14 constexpr covers nearly the whole
 48 interface.

49 Each type is a struct of two `std::uint64_t` words ordered by target endianness, so the object
 50 representation and alignment match a native 128-bit integer. Keeping both words unsigned
 51 lets the compiler treat the pair as one wide access, so loops over these types vectorize rather
 52 than scalarize. Operators forward to a native 128-bit type or intrinsic where one exists,
 53 and follow Knuth (Knuth, 1998) and Warren (Warren, 2013) over the two words elsewhere,
 54 with identical results. The library also supplies the interfaces the extension lacks, among
 55 them `<limits>`, `<bit>`, `<charconv>`, `<format>`, `<iostream>`, hashing, `Boost.Random` and
 56 `Boost.Math` integration, saturating arithmetic, the C23 checked-arithmetic interface (ISO/IEC,
 57 2024), and the `div_*` family of P3724 (Schultke, 2025).

58 Continuous integration runs the test suite natively on Linux, macOS, and Windows across
 59 `x86_64`, `x86_32`, `aarch64`, `ARM32v7`, `ARM64`, and big-endian `s390x`, plus `PPC64LE` under
 60 QEMU, using GCC 5 and later, Clang 5 and later, Visual Studio 2017 and later, Intel oneAPI
 61 DPC++, and NVCC. Because half of these targets have no native 128-bit type, the software
 62 fallbacks are tested as thoroughly as the intrinsic paths. GPU support is expressed through
 63 one annotation macro, `BOOST_INT128_HOST_DEVICE`, which expands to `__host__ __device__`
 64 under `nvcc` and `SYCL_EXTERNAL` under a SYCL compiler. More than eighty CUDA and eighty
 65 SYCL tests compare device results against the host, so agreement across that boundary is
 66 tested rather than assumed.

67 Where a native type exists, `Boost.Int128` matches or beats it: with GCC 14.2 on Linux `x86_64`,
 68 `uint128` division and modulo are roughly 16 percent faster than unsigned `__int128`, and
 69 signed multiplication is 1.5 times faster on big-endian `s390x`. Where no native type exists, the
 70 margins are larger: signed multiplication is 1.7 times faster than `std::Signed128` on MSVC
 71 `x86_64` and 6.2 times faster on MSVC `ARM64`, and 32-bit `x86` addition is 8 times faster than
 72 `Boost.Multiprecision`. A handful of cells favor an alternative, so the complete tables, rerun
 73 on every pull request, are published rather than a summary, at https://develop.int128.cpp.al/int128/u128_benchmarks.html and https://develop.int128.cpp.al/int128/i128_benchmarks.html.
 74 https://develop.int128.cpp.al/int128/i128_benchmarks.html.
 75 https://develop.int128.cpp.al/int128/i128_benchmarks.html.

76 Figure 1 shows the difference between a 64-bit and a 128-bit product.

```
#include <boost/int128.hpp>
#include <cstdint>
#include <iostream>

int main()
{
    using boost::int128::uint128;

    // Two values that each fit in 64 bits, but whose product does not
    constexpr std::uint64_t a {11400714819323198485ULL};
    constexpr std::uint64_t b {14029467366897019727ULL};

    // A 64-bit product silently wraps modulo 2^64
```

```
std::cout << " 64-bit: a * b = " << a * b << std::endl;

// The same multiplication is exact when the result is 128 bits wide
constexpr uint128 exact {uint128{a} * uint128{b}};
std::cout << "128-bit: a * b = " << exact << std::endl;

// Like a built-in integer, the types work in constant expressions
static_assert(exact / uint128{b} == uint128{a}, "Recovers a exactly");
static_assert(sizeof(uint128) == 16, "Exactly 128 bits on every platform");

return 0;
}
```

64-bit: a * b = 17693923505768731003
128-bit: a * b = 159945956516994065446882290339781513595

77 *Figure 1: The 64-bit product wraps modulo 2^{64} while the uint128 product is exact and*
78 *compile-time checked. The lower block is the program output.*

79 Research Impact Statement

80 As a component distributed through the Boost libraries, Boost.int128 is trivially in the reach
81 of researchers everywhere.

82 Boost.int128 serves as the arithmetic backend to the Boost Decimal Library (Borland &
83 Kormanyos, 2026). Using Boost.int128 is what allowed Boost.Decimal to be portable beyond
84 the platforms served by competing libraries, for example, on Apple's ARM64. It also allowed
85 portability to CUDA, enabling quantitative finance researchers to process data at much larger
86 scales than previously available.

87 The Boost Graph Library (Siek et al., 2002) already relies on 128-bit integers for exact geometry,
88 using int128 in is_straight_line_drawing to compute overflow-free orientation determinants,
89 showing that a dedicated Boost.Int128 would directly serve computational-geometry and
90 planarity research with applications in VLSI circuit layout, graph visualization, network/map
91 drawing, and GIS.

92 AI usage disclosure

93 Generative AI tools were used in developing this library and preparing this manuscript, for
94 drafting and editing prose, platform-specific optimizations, and generating test cases. All
95 library code, benchmark results, and text were reviewed and verified by the author, who takes
96 full responsibility for their correctness.

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